

HAI AI Index Report 2025

Summary

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The 2025 AI Index Report, produced by Stanford University's Institute for Human-Centered Artificial Intelligence (HAI), is the eighth edition of an annual comprehensive assessment of the state of artificial intelligence worldwide. Its purpose is to equip policymakers, researchers, industry leaders, journalists, and the public with rigorously validated, data-driven insights into AI's development, deployment, and societal impact. The report spans eight major domains: research and development, technical performance, responsible AI, economy, science and medicine, policy and governance, education, and public opinion.

Among the report's most significant findings is the extraordinary pace of AI advancement across nearly every dimension measured. AI performance on newly introduced benchmarks improved dramatically within a single year, with gains of up to 67 percentage points on coding tasks. The cost of AI inference plummeted over 280-fold in just 18 months, while hardware costs declined 30% annually and energy efficiency improved 40% per year. Open-weight models have nearly closed the performance gap with proprietary systems, and Chinese models have reached near-parity with American counterparts on major benchmarks, despite the United States maintaining its lead in producing notable AI models and attracting private investment—\$109.1 billion in 2024, roughly twelve times China's total.

The report documents AI's rapid embedding into everyday life and business. Organizational AI adoption surged from 55% to 78% in a single year, the FDA approved 223 AI-enabled medical devices by 2023, and autonomous vehicle services now provide over 150,000 rides weekly. Research consistently shows AI boosts worker productivity and narrows skill gaps. Meanwhile, AI earned unprecedented scientific recognition, with Nobel Prizes awarded for deep learning and protein folding research, and the Turing Award honoring reinforcement learning.

However, the report identifies critical challenges. Carbon emissions from training are escalating sharply, with recent models producing thousands of tons of CO₂. Responsible AI practices remain uneven: AI-related incidents rose 56% in 2024, trust in AI companies' data stewardship is declining, and implicit biases persist in leading models despite explicit mitigation efforts. Complex reasoning and planning remain unsolved problems, limiting AI's reliability in high-stakes applications. The data commons is also shrinking as websites restrict scraping for AI training.

Governments worldwide are responding with increased regulation and investment—U.S. federal AI regulations more than doubled in 2024, legislative mentions of AI rose ninefold since 2016, and countries collectively pledged hundreds of billions of dollars for AI infrastructure. Public opinion is cautiously optimistic globally, though deep regional divides persist, with developing nations far more enthusiastic than Western populations.

The report concludes that AI has moved decisively from experimental technology to a central force reshaping science, industry, and governance, making informed, evidence-based decision-making about its trajectory more essential than ever.